



Document Management System

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Child Health
	Code: CH-PICU-POL-3-Vers.1.0
	Effective Date: 12/10/2023 Target Review Date: 10/10/2026
	Title: Infection Control & Isolation Policy for PICU

1.0 Objective:

1.1 To preventtransmission of organisms between health care providers, patients and theirfamilies.

2.0 PolicyStatement:

2.1 In order toaccomplish control on infection in health care system, Royal Hospital hascertain recommendations for patient and health care workers safety.

2.2 Using theserecommendations this policy is formulated for isolation in PICU.

3.0 PolicyScope:

3.1 This policyapplies to all Healthcare workers (HCWs) who have direct contact with patientcare, which include physicians, nurses, emergency personnel, students, medicaland nursing students, pharmacists, hospital volunteers, housekeeping andadministrative staff and patients' family members.

4.0 Definitions:

4.1 **InfectionControl:** Refers to policies and procedures used to minimize the risk ofspreading infections, especially in hospitals and human or animal health carefacilities.

4.2 **Isolation:** SocialSeparation of a person who has or is suspected of having a contagious disease.

5.0 Equipment:

5.1 Gowns, gloves,masks, respirators, eye protection and teaching materials.

6.0 Procedure:

- HCWs are asked to attend staff clinic at the time of joining work to screen for HIV, HCV, HBV and Varicella Antibodies.
- All HCWs are asked to get N95 mask checked for appropriate size and fitting.
- A card will be provided to all HCWs which will have documented date of vaccinations.
- This card is asked to be carried in hospital on daily basis and to be displayed when ever asked.

6.2 Use dedicated stethoscope available near each bed for examination of patient.

6.3 Clean and disinfect reusable equipment including: IV pumps, cell phone and pagers (if used with in isolation area) or other electronics prior to removing from patient's room.

6.4 While entering patient area in PICU:

- Long sleeves are asked to be rolled up, white coats are discouraged and wearing scrubs is encouraged.
- All hand jewelry (e.g. rings, watches and bracelets) must be removed.
- Finger nail should be trimmed to minimum of 1/4th inch.
- If patient is isolated don PPE before entering isolation area, and discard before exiting the area.

6.5 There are two types of precautions:

6.5.1 Standard precautions:

- Include prudent preventive measures to be used at all times, regardless of a patient's infection status.
- Followed when caring for all patients. The effectiveness of simple protective barrier precautions reduces nosocomial colonization and infection.
- Most important component of this category is **HAND HYGIENE**. Hands are most common mode of transmission of organisms from one person to other, therefore following WHO's 5 moments for hand hygiene are very important and effective.
- Follow five moments of hand hygiene.

6.5.2 Transmission-based precautions:

There are two types of isolation in the ICU:

- Protective isolation for neutropenic or other immune compromised patients to reduce the chances of acquiring opportunistic infections.
- Source isolation of colonized or infected patients to minimize potential transmission to other patients or staff.

6.5.3 Airborne precautions:

- Disease-causing microorganisms may be suspended in the air as small particles, aerosols, or dust and remain infective over time and distance.
- Respiratory protection must be employed when entering the isolation room/area.

- For conditions, which need this type of isolation, refer to table given at the end.
- In immune-compromised host with varicella pneumonia, prolong duration of precautions for duration of illness.

6.5.4 Contact precautions:

- Infections can be spread by usual direct or indirect contact with an infected person, the surfaces or patient care items in the room.
- Isolation is required either in single room, cohort or side bed.
- Non-critical patient-care equipment should preferably be of single use. If unavoidable, then clean and disinfect them adequately before using to another patient.
- For conditions, which need this type of isolation, refer to table given at the end.

6.5.5 Droplet precautions:

- Microorganisms are also transmitted by droplets (large particles $>5 \mu\text{m}$ in size) generated during coughing, sneezing and talking, or a short-distance travelling.
- For conditions, which need this type of isolation, refer to table given at the end.
- Respiratory protection must be employed when entering the isolation room or within 3 feet of the patient.
- This should be worn by all persons entering the room, including visitors within 3 feet distance from patient.
- Isolation is required in single room or in cohort and distance between patients should be at least 1 meter or 3 feet.

6.5.6 Contact enteric precautions :

- For conditions, which need this type of isolation, refer to table given at the end.
- Isolation is required in single room.

7.0 Transporting Patients In Isolation

7.1 Contact and Contact Enteric Precautions:

- Clean your hands
- Put on gown and gloves to enter the room and transfer the patient to wheelchair or gurney
- Cover the patient from the neck down with a clean sheet or blanket
- Remove gown and gloves inside the patient's room and place in soiled linen hamper
- Clean your hands and put on clean gloves
- Transport patient
- Put on gown and gloves at destination if helping to transfer the patient
- Wipe down wheelchair or bed with hospital disinfectant

- Clean your hands

7.2 Droplet Precautions:

- Clean your hands
- Put on isolation and wear complete PPE if entering the patient's room
- Place isolation mask on the patient
- Remove your PPE inside the patient's room
- Clean your hands and transport the patient
- If patient is too young to wear a mask or cannot tolerate a mask: Wear your mask for the duration of the transport. Transport quickly to destination, maintaining a 3 foot distance from individuals along the route. This may require asking others to vacate elevators or confined spaces.

7.3 Suspected patients with infections in whom NPA is performed needs to get immediately isolated and continue isolation according to result.

7.4 To discontinue isolation refer to Table given at the end.

7.5 Waste product in isolated patient needs to get disposed in yellow bags.

7.6 For cleaning for of isolation area, refer to cleaning policy.

TYPE AND DURATION OF PRECAUTIONS RECOMMENDED FOR SELECTED INFECTIONS AND CONDITIONS

Infection/condition	Precaution		
	Type*	Duration†	Comment
Gastroenteritis	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks for gastroenteritis caused by all of the agents below
Adenovirus	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
Campylobacter species	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks

Cholera (Vibrio cholerae)	S		for the duration of illness or to control institutional outbreaks
C. difficile	C	DI	Discontinue antibiotics if appropriate. Do not share electronic thermometers; ensure consistent environmental cleaning and disinfection. Hypochlorite solutions may be required for cleaning if transmission continues. Hand washing with soap and water preferred because of the absence of sporicidal activity of alcohol in water less antiseptic hand rubs.
Cryptosporidium species	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
E. coli	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
Giardia lamblia	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
Rotavirus	C	DI	Ensure consistent environmental cleaning and disinfection and frequent removal of soiled diapers. Prolonged shedding may occur in both immunocompetent and immunocompromised children.
Salmonella species (including S. typhi)	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
Shigella species (Bacillary dysentery)	S		Use Contact Precautions for diapered or incontinent persons for the duration of illness or to control institutional outbreaks
<u>Hepatitis, viral</u>			
Type A	S		Provide hepatitis A vaccine post-exposure as recommended
Diapered or incontinent patients	C		Maintain Contact Precautions in infants and children <3 years of age for duration of hospitalization; for children 3-14 yrs. of age for 2 weeks after onset of symptoms.

Type B-HBSAg positive; acute or chronic		hemodialysis units	
Type C and other unspecified non-A, non-B		See specific recommendations for care of patients in hemodialysis units	Type of Precautions: A, Airborne; C, Contact; D, Droplet; S, Standard; when

A, C, and D are specified, also use S.

† Duration of precautions:

- CN, until off antimicrobial treatment and culture-negative;
- DI, duration of illness (with wound lesions, DI means until wounds stop draining);
- DE, until environment completely decontaminated;
- U, until time specified in hours (hrs) after initiation of effective therapy;
- Unknown: criteria for establishing eradication of pathogen has not been determined

Type and Duration of Precautions Recommended for Selected Infections and Conditions

Infection/condition	Precaution		
	Type*	Duration †	Comment
Herpes simplex			
Encephalitis	S		
Mucocutaneous, disseminated primary, severe	or C	Until lesions dry and crusted	
Mucocutaneous, recurrent (skin, oral, genital)	S		

Neonatal	C	Until lesions dry and crusted	vaginally or by C- section and if mother has active infection and membranes have been ruptured for more than 4 to 6 hrs until infant surface cultures obtained at 24-36 hrs. of age negative after 48 hrs incubation
<u>Herpes zoster (varicella-zoster) (shingles)</u>			
-			
-			
Disseminated disease in any patient			Susceptible HCWs should not enter room if immune caregivers are available; no recommendation for protection of immune HCWs; no
Localized disease in immunocompromised patient until disseminated infection ruled out	A,C	DI	recommendation for type of protection, i.e. surgical mask or respirator; for susceptible HCWs
Localized in patient with intact immune system with lesions that can be contained /covered		DI	Susceptible HCWs should not provide direct patient care when other immune caregivers are available.
HIV	S		Post-exposure chemoprophylaxis for some blood exposures
<u>Influenza</u>			
Human influenza	D	See Comments	Precaution for suspected or confirmed 7 days after illness onset or until 24 hours after the resolution of fever and respiratory symptoms, whichever is longer, while a patient is in a healthcare facility. May require longer periods based on clinical judgment, such as in the case of young children or severely immunocompromised patients, who may shed influenza virus for longer periods of time.
Pandemic influenza	D	See comments	Precaution for suspected or confirmed 7 days after illness onset or until 24 hours after the resolution of fever and respiratory symptoms, whichever is longer, while a patient is in a healthcare facility. May require longer periods based on clinical judgment, such as in the case of young children or severely

Type and Duration of Precautions Recommended for Selected Infections and Conditions

Infection/condition	Precaution		
	Type*	Duration †	Comment
Meningitis			
Aseptic (nonbacterial or viral; also see enteroviral infections)	S		Contact for infants and young children
Bacterial, gram-negative enteric, in neonates			
Fungal	S		
Haemophilus influenzae, type b known or suspected		U 24 hrs	
Neisseria meningitidis (meningococcal) known or suspected		U 24 hrs	See meningococcal disease below
Streptococcus pneumoniae	S		
Meningococcal disease: sepsis, pneumonia, meningitis		U 24 hrs	Post exposure chemoprophylaxis for household contacts, HCWs exposed to respiratory secretions; post exposure vaccine only to control outbreaks
Multidrug-resistant organisms (MDROs), infection or colonization	S/C		

resistant pneumoniae)	S.		
Mycoplasma pneumonia	D	DI	
Parainfluenza virus infection, respiratory in infants and young children	C	DI	Viral shedding may be prolonged in immunosuppressed patients. Reliability of antigen testing to determine when to remove patients with prolonged hospitalizations from Contact Precautions uncertain.

Type and Duration of Precautions Recommended for Selected Infections and Conditions

Infection/condition	Precaution		
	Type*	Duration †	Comment
Parvovirus B19 (Erythema infectiosum)	D		Maintain precautions for duration of hospitalization when chronic disease occurs in an immune-compromised patient. For patients with transient aplastic crisis or red-cell crisis, maintain precautions for 7 days. Duration of precautions for immunosuppressed patients with persistently positive PCR not defined, but transmission has occurred.
Pertussis (whooping cough)	D	U 5 days	Single patient room preferred. Post-exposure chemoprophylaxis for household contacts and HCWs with prolonged exposure to respiratory secretions
Pneumonia			
Adenovirus	D, C	DI	Outbreaks in pediatric and institutional settings reported. In immune-compromised hosts, extend duration of Droplet and Contact Precautions due to prolonged shedding of virus
Chlamydia	S		
Fungal	S		

Meningococcal	D	U 24 hrs	See meningococcal disease above
Mycoplasma (primary atypical pneumonia)	D	DI	
Streptococcus, group A	D	U 24 hrs	Contact Precautions if skin lesions present
Respiratory syncytial virus infection, in infants, young children	C	DI	Wear mask according to Standard Precautions. In immune-compromised patients, extend the duration of Contact Precautions due to prolonged shedding. Reliability of antigen testing to determine when to remove patients with prolonged hospitalizations from Contact Precautions uncertain.
Rhinovirus	D	DI	Droplet most important route of transmission. Outbreaks have occurred in NICUs and LTCFs. Add Contact Precautions if copious moist secretions and close contact likely to occur.
Severe acute respiratory syndrome (SARS)	A, D, C	DI plus 10 days after resolution of fever, provided respiratory symptoms are absent or improving	Airborne Precautions preferred; D if AIIR unavailable. N95 or higher respiratory protection; surgical mask if N95 unavailable; eye protection (goggles, face shield); aerosol-generating procedures and “super shedders” highest risk for transmission via small droplet nuclei and large droplets
Toxic shock syndrome	S		Droplet Precautions for the first 24 hours after implementation of antibiotic therapy if Group A streptococcus is a likely etiology
Viral hemorrhagic fevers	S, D, C	DI	Single-patient room preferred. Emphasize: 1) use of sharps safety

Addition to Standard Precautions Pending Confirmation of Diagnosis*

Clinical Syndrome or Condition†	Potential Pathogens‡	Empiric Precautions (Always includes Standard Precautions)
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DIARRHEA

Acute diarrhea with a likely infectious cause in an incontinent or diapered patient	Enteric pathogens§	Contact Precautions (pediatrics and adult)
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MENINGITIS

	<i>Neisseria meningitidis</i>	Droplet Precautions for first 24 hrs of antimicrobial therapy; mask and face protection for intubation
	Enteroviruses	Contact Precautions for infants and children
	<i>M. tuberculosis</i>	Airborne Precautions if pulmonary infiltrate Airborne Precautions plus Contact Precautions if potentially infectious draining body fluid present

RASH OR EXANTHEMS, GENERALIZED, ETIOLOGY UNKNOWN

Petechial/ecchymotic with fever (general) - If positive history of travel to an area with an ongoing outbreak of VHF in the 10 days before onset of fever	<i>Neisseria meningitidis</i> Ebola, Lassa, Marburg viruses	Droplet Precautions for first 24 hrs of antimicrobial therapy Droplet Precautions plus Contact Precautions, with face/eye protection, emphasizing safety sharps and barrier precautions when blood exposure likely. Use N95 or higher respiratory protection when aerosol-generating procedure performed
Vesicular	Varicella-zoster, <i>herpes simplex</i> , variola (smallpox), vaccinia viruses	Airborne plus Contact Precautions; Contact Precautions only if <i>herpes simplex</i> , localized zoster in an immunocompetent host or vaccinia viruses most likely

cough, coryza and fever	Rubeola (measles) virus	Airborne Precautions
RESPIRATORY INFECTIONS		
Cough/fever/upper lobe pulmonary infiltrate in an HIV-negative patient or a patient at low risk for human immunodeficiency virus (HIV) infection	<i>M. tuberculosis</i> , Respiratory viruses, <i>S. pneumoniae</i> , <i>S. aureus</i> (MSSA or MRSA)	Airborne Precautions plus Contact precautions
Cough/fever/pulmonary infiltrate in any lung location in an HIV-infected patient or patient at high risk for HIV infection	<i>M. tuberculosis</i> , Respiratory viruses, <i>S. pneumoniae</i> , <i>S. aureus</i> (MSSA or MRSA)	Airborne Precautions plus Contact Precautions Use eye/face protection if aerosol-generating procedure performed or contact with respiratory secretions anticipated. If tuberculosis is unlikely and there are no AIIRs and/or respirators available, use Droplet Precautions instead of Airborne Precautions Tuberculosis more likely in HIV-infected individual than in HIV negative individual
Cough/fever/pulmonary infiltrate in any lung location in a patient with a history of recent travel (10-21 days) to countries with active outbreaks of SARS, avian influenza	<i>M. tuberculosis</i> , severe acute respiratory syndrome virus (SARS- CoV), avian influenza	Airborne plus Contact Precautions plus eye protection. If SARS and tuberculosis unlikely, use Droplet Precautions instead of Airborne Precautions.
Respiratory infections, particularly bronchiolitis and pneumonia, in infants and young children	Respiratory syncytial virus, parainfluenza virus, adenovirus, influenza virus, Human metapneumovirus	Contact plus Droplet Precautions; Droplet Precautions may be discontinued when adenovirus and influenza have been ruled out

Skin or Wound Infection

Abscess or draining wound that cannot be covered

Staphylococcus aureus (MSSA or MRSA), group A streptococcus

Add Droplet Precautions for the first 24 hours of appropriate antimicrobial therapy if invasive Group A streptococcal disease is suspected

* Infection control professionals should modify or adapt this table according to local conditions. To ensure that appropriate empiric precautions are implemented always, hospitals must have systems in place to evaluate patients routinely according to these criteria as part of their preadmission and admission care.

Reference:

† Patients with the syndromes or conditions listed below may present with atypical signs or symptoms (e.g; neonates and adults with pertussis may not have paroxysmal or severe cough). The clinician's index of suspicion should be guided by the prevalence of specific conditions in the community, as well as clinical judgment.

‡ The organisms listed under the column "Potential Pathogens" are not intended to represent the complete, or even most likely, diagnoses, but rather possible etiologic agents that require additional precautions beyond Standard Precautions until they can be ruled out.

§ These pathogens include enterohemorrhagic *Escherichia coli* O157:H7, *Shigella spp*, hepatitis A virus, noroviruses, rotavirus, *C. difficile*.

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Indian Journal of Critical Care Medicine	Y. Mehta, A. Gupta et al.	2014	149-163
CDC: Management of multidrug-resistant organisms in healthcare settings	Siegel J D et al	2006	1-59

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